Optimization and Simulation of Collaborative Networks for Sustainable Production and Transportation

Giacomo Liotta, Toshiya Kaihara, *Member, IEEE*, and Giuseppe Stecca

***Abstract*—Complex and delocalized manufacturing industries require high levels of integration between production and transportation in order to effectively implement lean and agile operations. There are, however, limitations in research and applications simultaneously embodying further sustainability dimensions. This paper presents a methodological framework based on optimization and simulation to integrate aggregate optimized plans for production and multimodal transportation with detailed dynamic distribution plans affected by demand uncertainty. The objective function of the optimization model considers supply, production, transportation, and CO_2 emission costs, as well as collaboration over the multimodal network. Bill-of-materials and capacity constraints are included. A feedback between simulation and optimization is used to plan requirements for materials and components. Computational experiments are based on realistic instances. Results demonstrate that the framework can be effectively used to analyze cost-CO_2 emission tradeoffs, effects of demand uncertainty, and collaborative distribution strategies on economic and environmental performance of the supply chain.**

***Index Terms*—Computer simulation, mathematical programming, production planning, supply chain management.**

I. Introduction

MANUFACTURING networks nowadays consist of globally distributed production sites linked by material, information, and financial flows. Increased product complexity, shorter product lifecycles, and uncertainty in demand structure challenge multiple actors interacting in supply chain networks to achieve cost targets. However, the raising importance of sustainable manufacturing is changing strategies of single companies. Operationalization of sustainable practices throughout the supply chain is more difficult due to the multiple suppliers of products and services involved, and lack of practical approaches to concurrently deal with cost targets, demand uncertainty, and impacts on environment and society. Transportation operations are particularly relevant: they are carried out across several distance scales and must meet service level requirements of original equipment manufacturers (OEMs), suppliers, and customers. Depending on product types, logistics, and transportation may represent a relevant share of the total cost. Logistics and transportation, however, consist of essential processes to link manufacturers and enable them to operate according to push- and pull-based systems or combinations of both. Additionally, environmental impacts of transport constantly contribute to greenhouse-gas emissions increase. Literature highlights few works partly addressing integrated production and transportation including sustainability concerns. The objective of this work is to propose a methodological framework consisting of integrated quantitative methods to pursue the economic and environmental sustainability of networked production and transportation networks within the context of uncertain supply chains. This work analyzes the effects of uncertainty on both production and freight transport. Collaboration strategies are then tested in multimodal distribution as potential ways to concurrently strengthen economic and environmental performances. The adopted methods are multistage optimization and simulation to cover different system complexities, e.g., Bill-of-materials (BOM) constraints, regional cost differences between suppliers, OEMs as well as demand variations, and organizational settings characterizing operations of carriers. A realistic supply-chain scenario related to the automotive industry is adopted for the computational experiments. Sensitivity of transport performance to demand uncertainty is observed. Collaborative settings produce positive effects depending on the scope of supply chain operations and sustainability dimension. A collaborative setting is a set of criteria, mechanisms, and decisions made to operate coalitions of supply-chain partners and improve performance.

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II. Literature Review

Several research aspects characterize the problems to be solved in sustainable production and transportation. An overview of the main, relevant research streams is presented.

A. Multicommodity and Multimodal Distribution Networks

Several authors analyzed organizational aspects and problem solving issues related to production and multimodal freight transport. A recent literature review of models for multimodal transportation planning [1] makes use of a decision level-based classification while emphasizing: 1) at strategic level, origin-destination transportation options; 2) at tactical level, network

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flow planning problems and service network design problems; and 3) at operational level, real-time planning for orders, reaction and adjustment to disturbances, resource management, and itinerary re-planning. In production–distribution planning, optimization approaches require the development of multiple performance measures in the objective functions, e.g., cost, service level, social and environmental impacts [2]. BOM constraints have to be considered in multilayer production networks in order to study the propagation of the planning. BOM in the mixed integer linear programming (MILP) formulations have been used [3], [4]. A few works consider integrated environmental sustainability as well as other impact parameters in the objective functions, together with transport routing issues [5].

B. Collaborative Transportation

Collaboration in supply chains is driven by the variety of actors involved in supply chain processes and the intrinsic process complexity. Collaborative transportation systems can also be an effective target to couple and exploit mutual benefits between collaborative networks and sustainability [6].

Horizontal cooperation in supply chains and carriers selection was demonstrated to represent a source of potential benefits in order to increase carriers' profitability or improve quality of service [7]. Moreover horizontal cooperation is proven to imply structural requirements [8]; in fact, carrier selection often requires multicriteria approaches including significant environmental and energy use concerns [9]. In these settings, optimization and simulation are methods largely used in supply chain problems. Mathematical programming is used in supply chain orchestration built on collaboration in the potted-plant industry [10]. Optimization for collaboration models in transport is investigated in the forest industry [11]. Negotiation aspects are addressed in [12] and [13]. Collaboration is also important when integrated with information technology (IT) systems to simplify flows in complex logistics scenarios such as intermodal international hubs [14].

C. Optimization and Simulation

Optimization and simulation have been applied to investigate collaborative settings between logistics operators and customers in supply chains [15] and carriers [16]. The literature mainly focuses on the analysis of integrated production and transportation to support strategic decisions. However, it often does not consider tactical and operational aspects, e.g., in terms of lead times, route selection. In mode choice and carrier selection problems, the use of simulation and the inclusion of environmental impacts are under-represented [9]. Demand driven production systems require feedback between supply and demand, due to the strong reliance on forecasting in the scheduling process, and simulation can be of high help in these cases [17]. Simulation-based optimization is used to face demand uncertainty by Jung *et al.* [18] who propose a monthly planning system able to face uncertainty. Their approach partially inspired the present work. The most common source of uncertainty in supply chains is demand, and hybrid approaches are becoming popular [19]. With respect to the reviewed literature, the present work concurrently considers optimization for aggregate and detailed planning of production and transportation across a network of activated nodes, and simulation for providing the optimization with a feedback on demand and transport operations. CO_2 emissions of freight transport are included in both methods.

III. Scenario and Problem Description

The manufacturing scenario under study consists of a production and transportation network. Multiple components are manufactured by a subnetwork of potential suppliers and shipped to assembly plants. Components can be assembled in a subnetwork of manufacturing plants to produce finished products according to their BOM. Finished products are then shipped toward multiple market destinations over a final distribution network. Shipments are carried out by multiple carriers through multimodal transport between suppliers, assembly plants, and final market destinations. Supply, assembly plants, and transportation services are capacitated. Production and transportation costs consist of fixed component (e.g., plant/infrastructure cost) and a variable components (i.e., cost per unit produced/shipped). Production and transportation costs are specific for each supply and assembly site activated for producing a certain product or component, and transport mode and activated transport operators. CO_2 emissions are computed throughout the transport operations and vary according to the selected mode. Demand and transport lead times are uncertain. A first aggregate planning phase (push-based) must then be followed by a pull-based execution. Transport operations can be carried out in a baseline scenario without any form of cooperation among the transport operators and a collaborative setting among them. The problem is to identify shipment plans and configuration of activated suppliers and manufacturers for each BOM component and finished product, and transport operators and mode over the integrated network, which simultaneously minimize production costs, transportation costs, and CO_2 emissions to satisfy the variable demand throughout the time.

IV. Methodological Approach

The methodological approach to the problem consists of the integrated, multistage use of optimization and simulation (Fig. 1). In the first stage (stage 1), optimization provides a solution to the aggregate planning problem to fulfill the demand from suppliers and assembly plants. More specifically, the optimizer elaborates on the supply chain network which concurrently considers BOM and related supply network, capacitated assembly plants, capacitated multimodal transport connections, and markets. The optimization output, i.e., optimal aggregated planning corresponding to demand estimates, is then passed to a subsequent discrete-event simulation phase. In the second stage (stage 2), simulation tests the realization of the plan of final distribution of products throughout time. During the simulation run, the actual deliveries to market are recorded. The actual deliveries are affected by uncertainty in actual demand levels to be satisfied according to the plan and transport lead times

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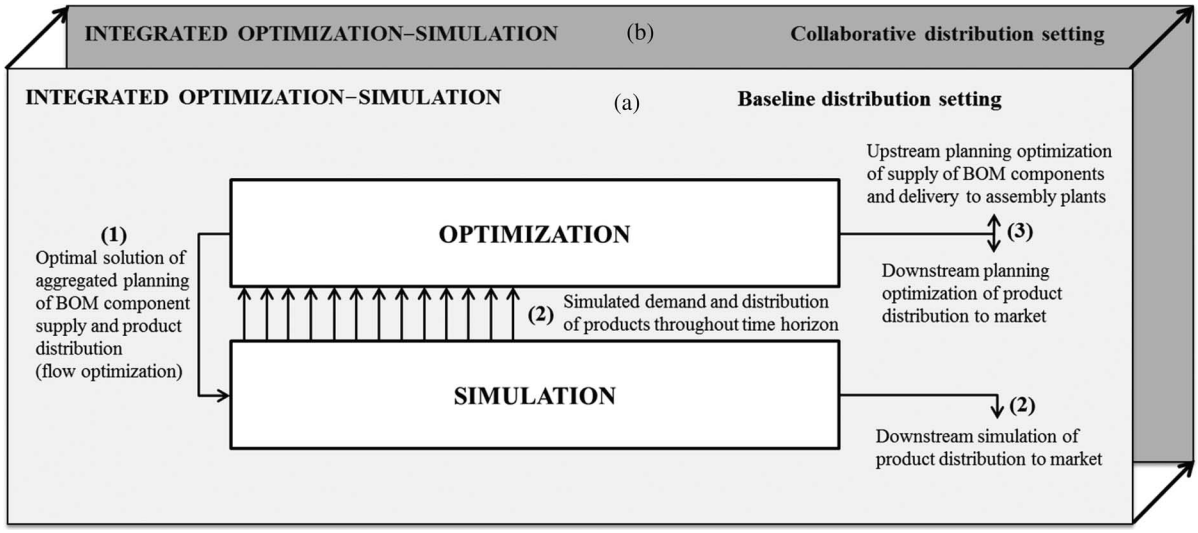


Fig. 1. Methodological approach: Integrated use of simulation and optimization under (a) baseline distribution settings and (b) collaborative distribution settings based on shared multimodal capacity.

depending on the transport modes. These actual deliveries data are provided to the optimization model while realizing a feedback between the simulator and the optimizer. In the third stage (stage 3), the optimization model carries out a further step to produce, according to the output dynamically recorded by the simulator, an updated plan for the upstream flows from suppliers to assembly plants and downstream flows from assembly to market demand areas.

The supply chain network then operates following a push-based (aggregated) planning in advance (stage 1) and a pull-based planned execution (stage 3) according to the simulation feedback (stage 2). Both optimization and simulation make use of key performance indicators such as total cost of production and transport as well as CO_2 emission levels. From the organizational viewpoint, this multistage approach is applied to two transport network settings: 1) baseline (traditional one) in which the activated transport service providers operate without any form of collaboration; and 2) collaborative setting in which the activated transport operators form coalitions and share the transport capacity of each available transport mode. The integrated optimization–simulation approach aims to strengthen the solution of the optimization model while integrating in a static solution its dynamic test through a feedback in order to optimally replan the supply, production, and final distribution. Furthermore, simulation allows including randomness, additional system parameters, and performance indicators such as, e.g., CO_2 emissions for each transport mode, capacity utilization, queues and waiting times of products, which increase the complexity of the optimizer. Potential benefits of the approach are then related to overcoming the limitations of "single-point-in-time" decisions obtained through static and deterministic optimization, and integrating optimal strategic and/or tactical planning (stage 1) with tactical/operational planning (stage 3 enabled by stage 2). Further benefits derive from the elaboration by simulation of an optimal solution as input and the use of simulation estimates of multiple performance indicators for production, logistics, and transportation. These estimates are dynamically obtained through interdependencies that could be hidden in an optimization. Thus, they can help improving the optimization model (e.g., through additional parameters or constraints) to better fit the specific industrial problem. On the other hand, the approach has the following limitations.

1) The optimization model would entail a coordinated decision-making process which is possible in presence of at least a collaborative setting (e.g., between an OEM and a transport operator), potentially with highly complex mechanisms.
2) Discrete-event simulation requires detailed modeling of system components, and assumptions have to be made while potentially entailing difficult model verification and validation.

V. Optimization Model

The optimization model considers several aspects of the production and distribution network. The aim is to find a global optimized solution for delocalized production considering fixed and variable production and distribution costs, and transport emission costs. The model considers BOM constraints in order to connect source of materials, component and final product assembly, and their related transportation. In stage 1, the model works for amounts aggregated yearly and it is not multiperiod. Dynamicity of demand and final product distribution network are delegated to the simulation model. After the simulation, the optimization model can be solved over monthly data (stage 3) in order to define in detail requirements for components as a material requirement planning tool. Multimodality is modeled as a multicommodity flow problem over a graph $G = (N, A)$ as done in [20]. BOM constraints and environmental sustainability considerations are added as in [21]. Moreover, the ability for carriers operating shipments to be aggregated in coalitions is modeled in this work. This feature was preliminarily taken into account in [16]. For each transportation service, defined as the connection $(i, j) \in A$, more than a carrier can operate the service. The carriers are grouped in different sets which

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define the coalitions that can be selected to operate the service. The parameter b_{lm} defines the coalition being equal to 1 if the carrier $l \in L$ is inserted in the coalition $m \in M$, 0 otherwise. The value of the binary decision variable y_m is 1 if coalition m is selected. The ability of carriers to operate on arcs is defined through the parameter a_{ijl}, equal to 1 if the carrier l can operate on the arc (i, j), 0 otherwise. The problem is to minimize the total production and transportation costs expressed as the sum of production, transportation, emissions (measured in quantity of CO_2), and costs related to coalition setup. For each node, $i \in N$ let $\Delta^+(i)$ the nodes $j \in N$ such that $(i, j) \in A$ and $\Delta^-(i)$, the nodes $j \in N$ such that $(j, i) \in A$ and K be the set of products (material, components, and final products). The subset $\bar{K} \subseteq K$ defines all the final products to be shipped to the demand areas. [*ED*, *DD*] is the earliness and lateness of the problem and bounds the total lead time of the supply chain. Let, moreover, s_i be the service time at node i, for each $i \in N$; t_{ijk} the time needed to traverse an arc $(i, j) \in A$ for shipment k; lts_k is the expression of inventory in time units. The k shipments are linked due to the BOM structural constraints. To this aim, we define a set $B \subset K \times K$ containing the precedence constraints between shipments which define the BOM structure. A tuple $(h, k) \in B$ defines the shipment h as a requirement (a material or a subassembly) for the shipment k (a final product or a subassembly). The model considers alternative sources for final products. To each shipment (final product) $k \in K$, it is associated a subset of alternative origin nodes $O^k \subset N$ in which the shipment can be produced with $O = \cup O^k$. D^k, $k \in K$, is the set of destination nodes for shipment k, while q_d is the quantity of shipment k to be shipped in destination $d \in D^k$. I^k are the subsets of intermediate nodes for shipment k with $I = \cup I^k$. The model considers the following cost components: fc_m, the fixed coalition setup cost for coalition m; fr_{ij}, the fixed routing cost for arc $(i, j) \in A$; c_{ij}, the transportation price required per unit of shipments transported along arc $(i, j) \in A$; and α, the cost of a tonne (t) of CO_2 emission. Capacity parameters are: C_{ij}, the capacity of arcs $(i, j) \in A$; and CP_{ik}, the production capacity of node i for product k. Other parameters are: *BigM*, a large constant; wsu_k, the weight of a unit of shipments of k in tonnes; and e_{ij}, the emission factors where $(i, j) \in A$ in tonnes of CO_2 for unit of shipment k. The variables of the problem are: y_m, $m \in M$, a binary variable defining whether the coalition m is selected ($y_m = 1$) or not ($y_m = 0$); u_{ijk}, $(i, j) \in A$, $k \in K$, a binary variable equal to 1 if shipment k is routed through arc (i, j), 0 otherwise; x_{ijk}, $(i, j) \in A$, $k \in K$, the quantity of shipment k routed through arc (i, j); w_{ik}, $i \in N$, $k \in K$, the arrival time at node i for shipment k. The model is formulated as follows:

$$\min \sum_{m \in M} \text{fc}_m y_m + \sum_{k \in K} \sum_{(i,j) \in A} \text{wsu}_k c_{ij} x_{ijk} + \sum_{k \in K} \sum_{(i,j) \in A} \text{fr}_{ijk} u_{ijk} + \sum_{k \in K} \sum_{i \in O^k} \sum_{j \in \Delta^+(i)} p_{ik} x_{ijk} + \sum_{k \in K} \sum_{i \in O^k} \sum_{j \in \Delta^+(i)} f_i z_{ik} + \alpha \sum_{(i,j) \in A} \sum_{k \in K} \text{wsu}_k e_{ij} x_{ijk} \quad (1)$$

subject to

$$x_{ijk} \leq \sum_{l \in L} \sum_{m \in M} a_{ijl} b_{lm} y_m BigM \quad \forall (i, j) \in A, k \in K \quad (2)$$

$$\sum_{m \in M} b_{lm} y_m \leq 1 \quad \forall l \in L \quad (3)$$

$$\sum_{i \in O(k)} \sum_{j \in \Delta^+(i)} x_{ijk} = \sum_{d \in D^k} q_d \quad \forall k \in \bar{K} \quad (4)$$

$$\sum_{i \in \Delta^-(d)} x_{idk} = q_d \quad \forall d \in D^k,\ k \in \bar{K} \quad (5)$$

$$\sum_{i \in O^k} \sum_{j \in \Delta^+(i)} x_{ijk} - \sum_{j \in D^k} \sum_{i \in \Delta^-(j)} x_{ijk} = 0 \quad \forall k \in K/\bar{K} \quad (5a)$$

$$\sum_{j \in \Delta^+(i)} x_{ijk} - \sum_{j \in \Delta^-(i)} x_{jik} = 0 \quad \forall k \in K\ \forall i \in I^k \quad (6)$$

$$\sum_{j \in \Delta^-(i)} x_{jih} - \sum_{(h,k) \in B} \sum_{j \in \Delta^+(i)} x_{ijk} = 0 \quad \forall i \in O, h \in K/\overline{K} \quad (6a)$$

$$w_{ik} + s_i + t_{ijk} - w_{jk} \leq (1 - u_{ijk}) BigM \quad \forall k \in K\ \forall (i, j) \in A \quad (7)$$

$$w_{dh} + s_d + \text{lts}_k - w_{dk} \leq 0 \quad \forall (h, k) \in B,\ d \in D^h \quad (7a)$$

$$ED \leq w_{ik} \leq DD \quad \forall k \in K\ \forall i \in \{O, D\} \quad (8)$$

$$x_{ijk} \leq C_{ij} u_{ijk} \quad \forall (i, j) \in A, k \in K \quad (9)$$

$$\sum_{i \in O(k)} \sum_{j \in \Delta^+(i)} x_{ijk} \leq CP_{ik} \quad \forall i \in N\ \forall k \in K \quad (10)$$

$$x_{ijk} \geq 0, w_{ik} \geq 0, \quad u_{ijk} \in \{0, 1\}\ \forall k \in K\ \forall (i, j) \in A, \quad y_m \in \{0, 1\}\ \forall m \in M. \quad (11)$$

The objective function (1) consists of a first term considering the coalition setup costs. The second and third terms consider, respectively, variable and fixed transportation costs of each travelled arc; fourth and fifth terms consider variable and fixed production costs, while the sixth one the CO_2 emission cost. Constraint (2) links the flow variables to the service operated by the carriers of the selected coalitions. Constraint (3) imposes the carriers to belong to a single selected coalition. Constraints (4), (5), (5a), (6), and (6a) are classical flow constrains applied to multicommodity flow. In particular, (5) and (6) are applied to final products while (5a) and (6a) are applied to components and materials. Functions (7) and (8) define time constraints for products; function (7a) defines synchronization for required BOM materials; (8) is a time window constraint. Functions (9) and (10) define capacity constraints while function (11) the decision variables.

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TABLE I
FREIGHT TRANSPORT PARAMETERS

Transport services	Transport cost[a] (EUR/tkm)	CO_2 emissions[b] (gCO_2/tkm)	CO_2 emission cost[c] (EUR/tCO_2)
Road	0.14[d]	93.1[e]	11.07
Rail	0.11	17.4	11.07
Sea	0.009	101.0[f]	11.07

[a]W. Schade *et al.*, "COMPETE Final Report: Analysis of the contribution of transport policies to the competitiveness of the EU economy and comparison with the United States," Funded by European Commission—DG TREN. Karlsruhe, Germany, Oct. 2006.
[b]*CO_2 Information for Transport Services, Application of Article L. 1431-3 of the French Transport Code—Methodological Guide*, French Republic—Ministry of Ecology, Sustainable Development and Energy, and Environment and Energy Management Agency, Oct. 2012.
[c]*Turning the Corner: Regulatory Framework for Industrial Greenhouse Gas Emissions*, Government of Canada, Ottawa, Mar. 2008. (Value in the table converted from USD 15 to EUR).
[d]Heavy duty vehicles (HDV).
[e]Semitrailer truck with a GCW of 40 tonnes, large volumes, road diesel.
[f]Short sea shipping (Ro-Ro).

VI. DISCRETE EVENT SIMULATION MODEL

The simulation model reproduces the production and distribution from assembly plants to market destinations. The model mimics the following subsystems:

1) demand generation of final products according to the aggregated plan of the optimizer;
2) multimodal transportation operations including temporary inventories of products being shipped;
3) final market destinations.

A discrete event simulation model has been implemented and validated with the simulation software Simio which was used under a grant from Simio LLC. The planned demand is monthly generated and randomly varied by $\pm 10\%$ following a uniform probability density function. The sources of the discrete-event model calculate the fixed and variable production cost of each assembly plant. Three freight transport modes are considered: 1) road; 2) rail; and 3) sea. Transport capacities and daily frequencies for each transportation service and activated carrier (output of the optimization) are modeled. Transport lead times follow a triangular probability density function. Transport costs consist of fixed and variable components which differ between transport modes and organizational setting (baseline and collaborative distribution settings). Total transport costs are elaborated according to cost per tonne km (tkm) of each transport mode (transport operating costs in EU-25 listed in Table I). Accordingly, CO_2 emissions of transport are totally calculated as well as for each transport mode together with an associated cost (Table I). The simulation time horizon is 52 weeks. Simulation enacts the two final distribution settings: baseline and collaborative. In the baseline setting, a specific carrier is preferably associated to each assembly plant. In the collaborative setting, modal capacity is shared and allocated following the criteria of transport distance between origin (assembly plant) and destination (market area).

TABLE II
POTENTIAL COALITIONS DEFINED IN THE MATRIX $b[l][m]$

Carrier l	Coalition m — baseline							Coalition m — collaborative			
	1	2	3	4	5	6	7	1	2	3	4
1	1	0	0	0	0	0	0	1	0	1	0
2	0	1	0	0	0	0	0	1	0	1	0
3	0	0	1	0	0	0	0	0	1	1	0
4	0	0	0	1	0	0	0	1	0	1	0
5	0	0	0	0	1	0	0	0	1	1	1
6	0	0	0	0	0	1	0	1	0	1	1
7	0	0	0	0	0	0	1	0	1	1	0

VII. MODELING THE COLLABORATIVE DIMENSION

Causal mechanisms and drivers of collaboration in transportation can be framed into several perspectives, i.e., technical, risk, financial, organizational, and operational [22]. A collaborative setting concerns the definition of collaboration plans, allocation and sharing of resources, and benefits. The approach followed in this paper mainly addresses a tactical-operational perspective where the variables controlled for the selection of the proper collaboration setting are cost (i.e., coalition setup and transportation), transport capacity, and carbon footprint that are listed, among others, in [22]. In order to integrate the collaboration into the production–distribution network, at this stage of the research, other characteristics of collaborative settings are not included (e.g., profit sharing and low-level negotiation between partners of a coalition). The decisional framework considers the coalition setup cost (e.g., for integration of information systems) as a strategic variable for selecting the proper coalition. This variable is defined in Section V and denoted as y_m. The "coalitions" in the baseline scenario (without collaboration) contain one carrier only, thus providing the simple set of activated operators in this setting (Table II). In the collaborative setting, the potential coalitions to be selected are a predefined subset of possible combinations of carriers that can be *a priori* composed, e.g., on the basis of prenegotiated options (Table II). The optimization model selects the most convenient coalition(s), i.e., the minimum number of coalition(s) covering the network, by considering costs and emissions. The simulation model addresses the collaborative dimension by replicating the coalitions obtained in the optimization and implementing the shared mode capacity between transport operators (tactical dimension), as well as the distance ranges in the logic to select transport mode routes served by coalition members (operational dimension). The approach explicitly and implicitly models collaboration characteristics to generate the plans and analyze the system performance.

VIII. CASE STUDY AND SCENARIO SETTING DETAILS

The defined case study reflects a realistic scenario of the automotive industry over a European production and transportation network. General data have been retrieved through publicly available sources and elaborated according to the problem setting. Three final products, each one consisting of three main components, must be delivered to seven market

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TABLE III
OPTIMIZATION EXPERIMENTS

Cost (mEUR)	Stage 1 Baseline	Stage 1 Collab.	Stage 3 Baseline	Stage 3 Collab.
Total cost	1557.70	1557.30	1509.23	1508.55
Transport	144.20	143.80	176.31	177.90
Production	1412.02	1412.02	1327.53	1329.12
CO_2 emiss.	1.48	1.48	1.50	1.53

areas. Six suppliers and three assembly plants can be potentially activated to manufacture specific BOM components and finished products, respectively. Seven transport operators (carriers) can provide the services: three operators for the road transport, two operators for the rail transport, and two for the sea transport. Fixed production costs for activating in a selected site the manufacturing of a component vary in the range 1.8–4 mEUR while for a product in the range 9–12 mEUR. Component and finished product variable costs vary between 30 and 300 EUR/unit, and 1200–3400 EUR/unit, respectively. Transport costs and CO_2 emissions are presented in Table I. The aggregated demand for all the finished products is equal to 575 000 units/year. Each product and component is characterized by a specific weight (t). The baseline scenario does not consider any cooperation form in the integrated network. The simulation experiments test a further dimension of the collaborative settings (collaboratives 1 and 2) in the final distribution by considering two levels of distances to decide on the assignment of shipments to each transport mode: up to 700 and 1000 km, respectively (road mode); between 700 and 1400 km as well as between 1000 and 2000 km (rail), respectively; longer than 1400 km and longer than 2000 km, respectively (sea). For the highest level of distance ranges, transportation times and capacity of services are increased in order to take into account the longer distances and calibrate the model to reach a comparable level of products delivered to the market.

IX. COMPUTATIONAL RESULTS

The results of the optimization of the global network are presented in Table III. In the baseline scenario, only a subset of available operators is selected while in the collaborative scenario all the available transport operators are selected. The experiments show that the optimization of aggregated plans (stage 1) for both baseline and collaborative (collab.) scenario settings are substantially equivalent in terms of costs. Only a 0.2% reduction in transport cost can be noticed in the collaborative setting. The results of the optimization after the feedback from the simulation model (stage 3) are again aligned but a subtle difference emerges in the collaborative setting, e.g., -0.5% and 1.5% in total and emission costs, respectively. More interestingly, comparisons between the optimization stages 3 and 1 highlight that in both the settings.

1) The total and production costs decrease by approximately 3% and 6%, respectively.
2) The transport and CO_2 emission costs increase in the range 22%–24% and 2%–4%, respectively.

It must be observed that in stage 3 the optimizer sequentially processes the monthly demand randomly generated by the simulation (demand is generated throughout 13 periods

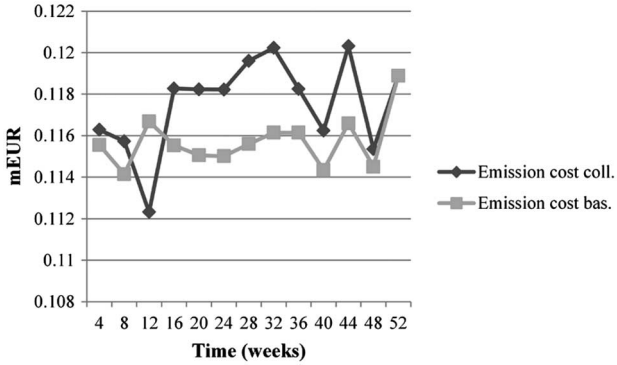


Fig. 2. Optimization (stage 3): CO_2 emission cost trends in the baseline and collaborative scenarios.

consisting of 4 weeks and each period represents a temporarily separated input for the optimization). The optimization is then carried out sequentially over 13 separated steps. The observed differences between the planned and simulated total demand in baseline and collaborative scenarios are in the range 0.2%–0.4%. On the one hand, the randomness effect on demand causes the mentioned reduction in production and total costs. This can be explained by reduced fixed costs due to a smaller number of production or supply sites activated to produce certain products/components. On the other hand, randomness more significantly impacts on transport costs and relatively on CO_2 emissions. This is related to longer distances between fewer assembly plants/supply sites activated for certain products as well as assembly plants and market destinations. The pull-based strategy has then negative effects on emissions due to the transport cost structure, especially upstream the assembly plants. Therefore, random variability of demand may more significantly affect both economic and environmental performances of freight transport. Figs. 2 and 3 show the trends of costs throughout the optimization steps. A higher variability can be noticed in the collaborative setting because of the activated operators for highly different transport modes.

By focusing on the distribution from assembly plants to final market destinations, the simulation experiments (stage 2 of the methodological approach) compare the results of the baseline setting (baseline) to the two collaborative settings based on sharing the modal transport capacity between carriers (Table IV). Simulations do not consider transport and emissions upstream the assembly plants which are considered in the static planning of the optimization model.

In the collaborative setting 1 (collaborative 1), an important decrease in average transport cost (including CO_2 emission costs) can be noticed. However, the CO_2 emission level increases by approximately 11% (see the percentage variation C1-B in Table IV). This result can be explained by the increased share of shipments carried out by the sea mode which is cheaper but has a higher environmental impact in this problem setting. In fact, both the throughput and tonne-km carried through the sea modality remarkably increase. Sea transport absorbs freight flows from the road mode in terms of tonne-km (-79.3%) which are carried for longer distances. More significantly for the environmental performance, sea transport captures tonne-km from the rail mode (-26.7%) which has the lowest environmental impact. Standard deviation (SD) and

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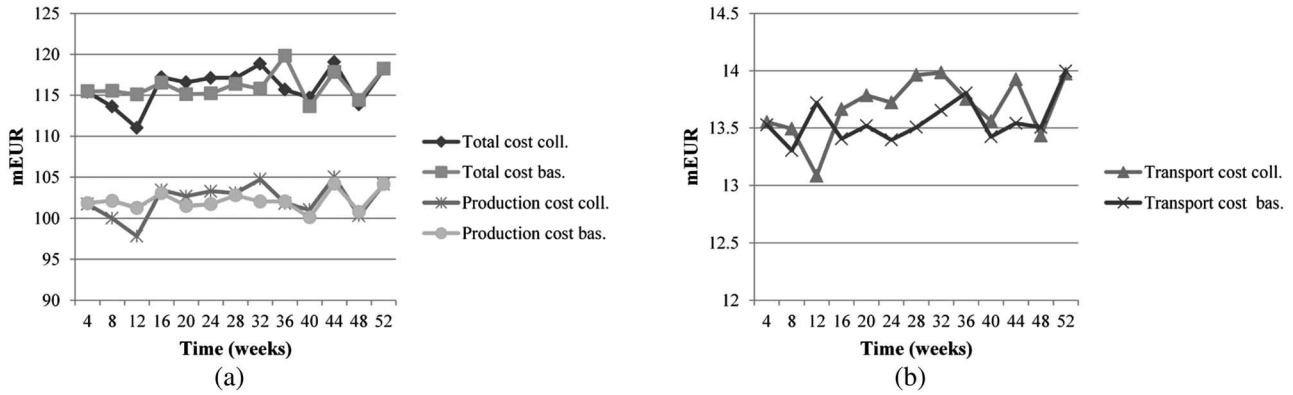


Fig. 3. Optimization (stage 3). Total cost and cost trends of (a) production and (b) transportation in the baseline and collaborative scenarios.

TABLE IV
SIMULATION EXPERIMENTS

Indicator[a]	Baseline (B)			Collaborative 1 (C1)			C1-B	Collaborative 2 (C2)			C2-B
	Average	SD	HW	Average	SD	HW	**Var. %**	Average	SD	HW	**Var. %**
Products delivered (t)	702, 173	2648	756	703, 416	2458	702	**0.2**	702, 138	2693	769	**0.005**
Production cost (mEUR)	1220.66	4.39	1.25	1222.38	4.24	1.21	**0.1**	1220.81	3.99	1.14	**0.01**
Transport cost (mEUR)	106.57	0.53	0.15	53.31	0.23	0.07	**–50.0**	80.61	0.35	0.10	**–24.4**
CO_2 emissions (t)	83, 432	392	112	92, 735	389	111	**11.1**	70, 498	395	113	**–15.5**
Road transport											
Throughput (t)	217, 191	1469	420	206, 556	1629	465	**–4.9**	231, 582	1321	377	**6.6**
Million tkm carried	444.98	3.20	0.91	92.03	0.70	0.20	**–79.3**	117.30	0.78	0.22	**–73.6**
Number in station (t)	438.8	20.0	5.7	258.7	16.8	4.8	**–41.0**	265.4	18.7	5.4	**–39.5**
Time in station (h)	16.8	0.9	0.3	10.2	0.7	0.2	**–39.6**	9.3	0.65	0.19	**–44.7**
Rail transport											
Throughput (t)	240, 210	1344	384	177, 310	1705	487	**–26.2**	298, 890	1901	543	**24.4**
Million tkm carried	283.90	2.05	0.59	208.00	2.02	0.58	**–26.7**	470.02	2.83	0.81	**65.6**
Number in station (t)	801.4	48.5	13.8	370.8	45.8	13.1	**–53.7**	641.5	71.7	20.5	**–20.0**
Time in station (h)	27.0	1.6	0.5	17.0	2.1	0.6	**–37.3**	17.4	1.96	0.56	**–35.7**
Sea transport											
Throughput (t)	244, 772	1753	501	319, 550	1544	441	**30.6**	171, 666	1265	361	**–29.9**
Million tkm carried	366.97	2.45	0.70	797.50	3.90	1.11	**117.3**	508.90	3.74	1.07	**38.7**
Number in station (t)	973.5	84.9	24.2	893.0	79.6	22.7	**–8.3**	632.4	72.6	20.7	**–35.0**
Time in station (h)	32.2	2.8	0.8	22.7	2.0	0.6	**–29.7**	29.9	3.45	0.99	**–7.3**

[a]Simulation results include SD and HW (95% confidence interval) for each setting that was tested by making 50 replications.

half-width (HW) values are very low. Furthermore, the structure of the demand, i.e., quantity and distance from assembly plants, affects this outcome in the collaborative setting. Remarkable improvements in transport service levels (products in the system and waiting times) can be noticed for all the transport modes, thus highlighting a streamlined flow of products throughout the distribution network. The average waiting time and product quantities in transport stations exhibit a more variable performance than other metrics in both settings (SD varies from approximately 5% to 10% in the baseline and from 6.5% to 12.3% in collaborative 1 settings). Scheduled capacity utilizations of transport modes vary in line with the throughput variations, e.g., the sea mode is more used in this collaborative scenario (+30.8%) than in the baseline.

The additional experiment on the collaborative setting (collaborative 2) aimed to analyze the effect of an increase in the thresholds of the distances to assign shipments to modal transport operators. It can be noticed that average transport costs decrease by approximately 24% but, in this case, the CO_2 emissions noticeably decrease as well, by 15.5% (Table IV, column C2-B). In this collaborative experiment, under substantially comparable levels of delivered products and scheduled utilization, the service levels improve together with the economic and environmental performance of freight transport. The increase in freight flows traveling across the rail modality predominantly contributes to the combined effect on costs and CO_2 emissions. The distance thresholds of this scenario impact on the shift of carried freight from sea to rail mode. SDs and HWs are similar to collaborative 1 results. The ameliorative collaboration settings produce positive effects on the final distribution. However, according to the simulations, criteria and calibration of parameters for decision-making (e.g., distance thresholds, CO_2 or transport costs for each modality) seem to be particularly relevant.

X. CONCLUDING REMARKS

This work contributes to the study of complex production and distribution systems by integrating the use of optimization and simulation, and investigating production and transportation options to improve economic and environmental performances. Computational results show that, in the overall production and

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transportation network, random variability of demand has a higher impact on freight transport performance in terms of increased costs and CO_2 emissions, whereas total and production costs decrease. Furthermore, in the final distribution, collaborative settings among carriers can simultaneously produce positive economic and environmental effects. In order to achieve this combined effect, it appears important to properly adjust the parameters of emissions and transport performance or to investigate which targets for them must be set to achieve a desired performance. The practical applicability of the proposed approach requires a supply chain in which coordinated planning, control and exchange of data are fully enabled. This would be possible in a supply chain which is controlled by a focal company or formed and governed by collaborative settings among partners, i.e., supplier, OEM, and logistics service provider (LSP). In both cases, at least a tight collaboration between OEM and LSP has to be established. This work can be applied to distribution planning in regions where specificities of suppliers and transport operators may be positively integrated in detailed production–distribution plans. Future research will refine and extend collaborative settings including additional complexities to enhance the collaboration mechanisms. Network modeling in optimization and simulation will also be addressed.

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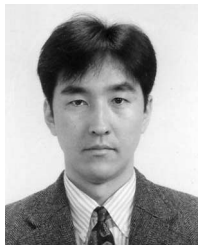
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